

What helps against chronic ankle instability?

Issuing institution: Institute for Quality and Efficiency in Health Care (IQWiG) / NCBI Bookshelf

Publication/review date: Updated 24 June 2022

Original official URL: <https://www.ncbi.nlm.nih.gov/books/NBK279551/>

Archive note: English text snapshot of the official patient-information webpage. Navigation was removed; the medical content was not translated.

Ankle sprains: Learn More – What helps against chronic ankle instability? - InformedHealth.org - NCBI Bookshelf

NCBI Bookshelf. A service of the National Library of Medicine, National Institutes of Health.

InformedHealth.org [Internet]. Cologne, Germany: Institute for Quality and Efficiency in Health Care (IQWiG); 2006-.

Ankle sprains: Learn More – What helps against chronic ankle instability?

Last Update:

June 24, 2022

; Next update: 2025.

Strength and coordination exercises help in the treatment of chronic ankle instability. But surgery is sometimes a good idea too. People who already start rehabilitation exercises two to three weeks after surgery will probably become active again sooner than people who only start later on.

People who have a sprained ankle might develop chronic (long-lasting) ankle instability. It is considered to be chronic if the

ankle joint

still gives way too easily six months after the first sprain, or if the ankle is sprained again within six months of the first sprain. Unstable ankle joints can easily be twisted again. The joint remains unstable in about 10 to 20 out of 100 people who have a severe ankle sprain.

This kind of instability can develop if the ankle ligaments are overstretched, or if they are torn and grow back together too loosely (mechanical instability).

The interactions between the bones in the ankle and the surrounding
ligaments

and muscles may have changed too. Our bodies have an awareness of their own movement and location, known as proprioception. This also helps to coordinate the movements of the

joints

, stabilize them using reflexes, and keep the body

balanced

. So if an ankle feels permanently unstable, this might not only be caused by overstretched ligaments, but also by a problem with proprioception or resulting problems with muscle coordination.

What treatment options are there?

There are various treatment options for chronic ankle instability: Conservative treatment is tried first. This involves doing physical therapy (physiotherapy) to strengthen the joint, and possibly wearing an ankle brace or splint to stabilize it.

One common

physical therapy

approach is called neuromuscular training. It aims to improve the ankle's strength, stability and coordination. Research has shown that neuromuscular training can actually improve the stability and the movement of the ankle in the first few weeks. But there isn't enough research on the long-term effectiveness of this treatment.

If the joint remains unstable despite this training because the ligaments are too loose,

surgery

may be considered. One option is to shorten and tighten the ankle ligaments. Another option is to take a tendon out of the lower leg and use it as an outer ligament on the ankle.

Tendons

and ligaments are made from similar tissue.

How can you get fit again after surgery?

Ankle instability often leads to problems with muscle coordination. For this reason, exercises are always an important part of rehabilitation after surgery. It also helps to wear an

ankle brace or splint

during this time, to give the ankle external support. The pressure it puts on the joint can help you to regain a good sense of muscle coordination too.

People probably benefit from starting movement, strength and coordination exercises two to three weeks after surgery at the latest. Studies suggest that people who do this become active again sooner than those keep their foot still for six weeks and don't do any exercises during that time. The study participants who had started doing exercises and strength training earlier were able to go back to work about one to two weeks sooner. They were also able to do sports again about three weeks sooner. But no advantages could be seen anymore after about two years: The stability and flexibility of participants' ankle joints were the same in both groups. Many of these studies had few participants, though, or were

poorly done

. This means that their results need to be interpreted with caution.

Which is better: physical therapy or surgery?

There were no studies comparing surgery directly with physical therapy or other treatments. For this reason, it's not possible to say who would benefit most from surgery, or how effective it is compared with non-surgical (conservative) treatment.

It is also not clear how the various surgical procedures compare with one another. There are only few small studies on this, and they don't provide reliable results. One study looked at a surgical procedure called "Chrisman Snook," in which the ligaments are strengthened using tendons taken from the foot or the lower leg. Compared to other procedures, it resulted in more complications, including

nerve damage

.

Overall, it's currently not clear whether surgery leads to a faster recovery than strength and coordination training does. But if the ankle remains unstable because of loose ligaments, surgery might be an option. Regardless of which treatment you choose: With a little patience, sprained ankles usually become stable again.

Sources

De Vries JS, Krips R, Sierevelt IN et al.

Interventions for treating chronic ankle instability

. Cochrane Database Syst Rev 2011; (8): CD004124. [

PMC free article

: PMC13254623

]]

PubMed

: 21833947

]

Powden CJ, Hoch JM, Hoch MC.

Rehabilitation and Improvement of Health-Related Quality-of-Life Detriments in Individuals with Chronic Ankle Instability: A Meta-Analysis

. J Athl Train 2017; 52(8): 753-765. [

PMC free article

: PMC5561776

][

PubMed

: 28704635

]

IQWiG health information is written with the aim of helping people understand the advantages and disadvantages of the main treatment options and health care services.

Because IQWiG is a German institute, some of the information provided here is specific to the German health care system. The suitability of any of the described options in an individual case can be determined by talking to a doctor. informedhealth.org can provide support for talks with doctors and other medical professionals, but cannot replace them. We do not offer individual consultations.

Our information is based on the results of good-quality studies. It is written by a team of health care professionals, scientists and editors, and reviewed by external experts. You can find a detailed description of how our health information is produced and updated in our methods.

© IQWiG (Institute for Quality and Efficiency in Health Care)

Bookshelf ID: NBK279551

Contents

< Prev

Next >